

State of Developer Ecosystem 2024: Language Trends, Satisfis

Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization

Usage shift

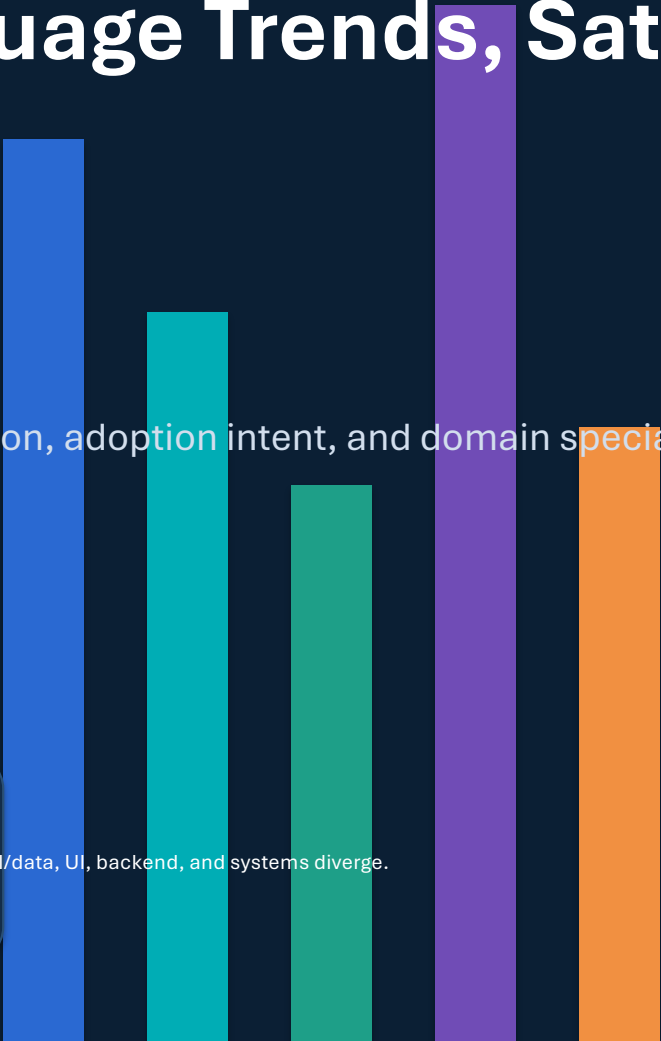
Python has risen from 32% to 57%, nearly closing the gap

Preference signal

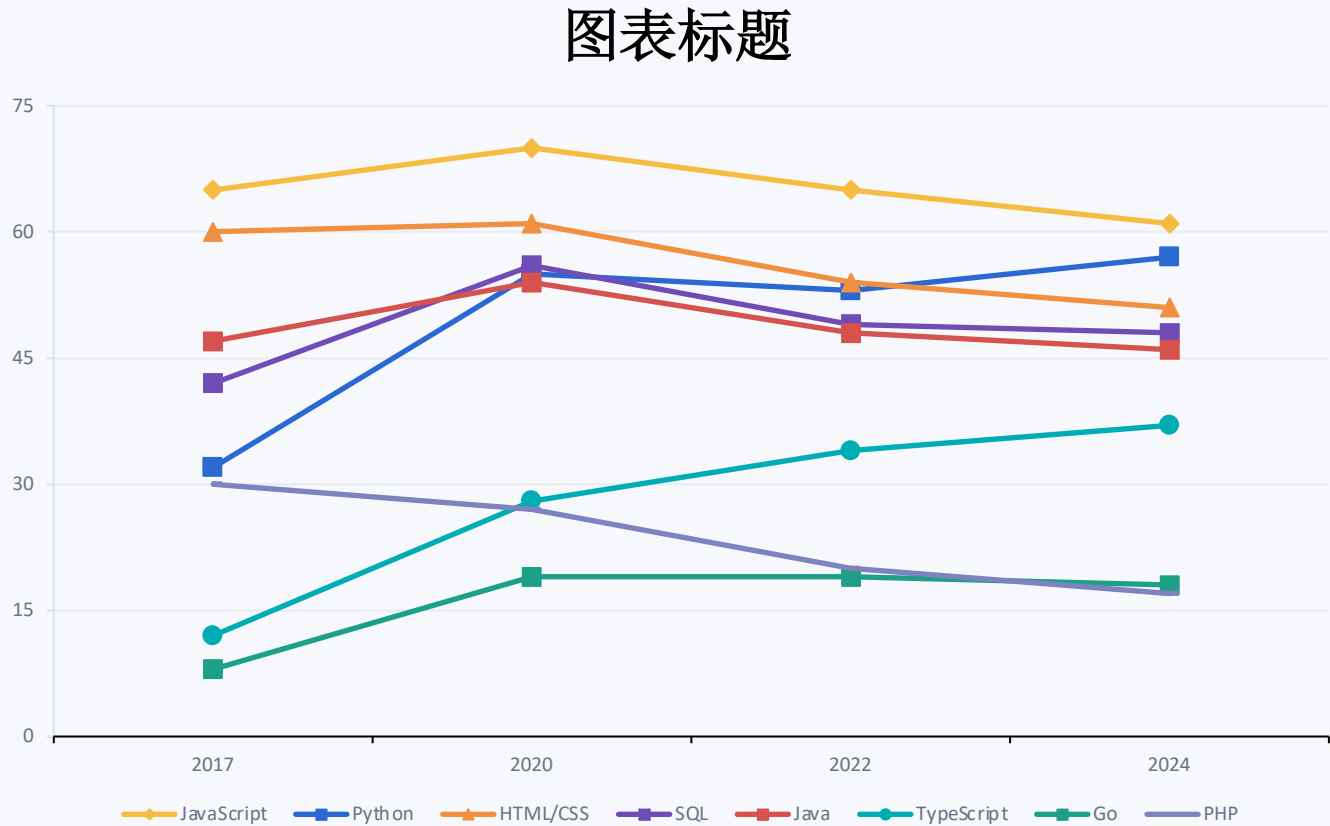
TypeScript and Rust lead satisfaction, reflecting demand f

Strategy lens

Language choice is increasingly domain-specific: AI/data, UI, backend, and systems diverge.



Python Closes the Gap: 8 Years of Language Usage Shifts



Comparison table: available survey points

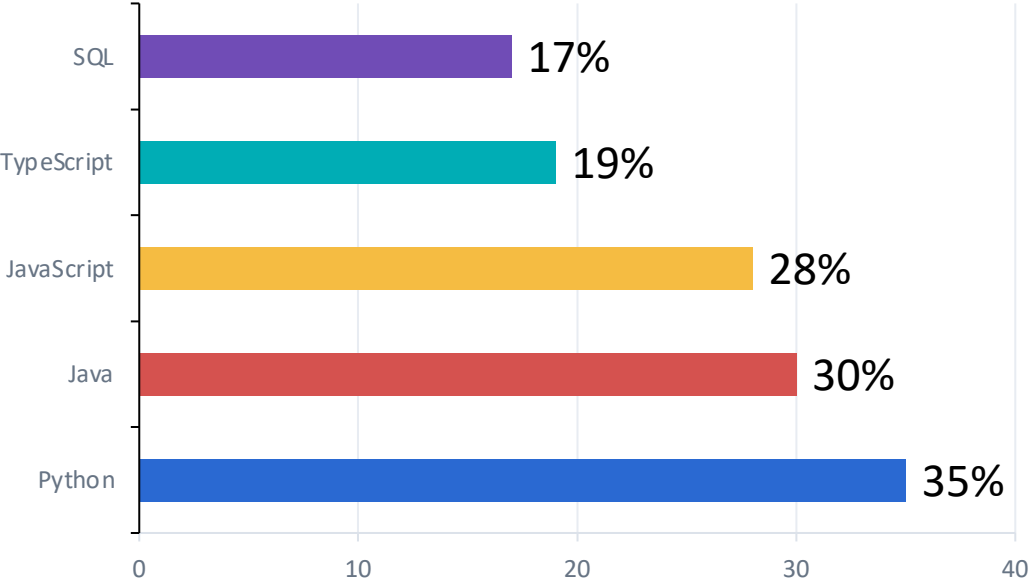
Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	Declining from peak
Python	32%	55%	53%	57%	Steady growth
HTML/CSS	60%	61%	54%	51%	Gradual decline
SQL	42%	56%	49%	48%	Plateaued
Java	47%	54%	48%	46%	Slow decline
TypeScript	12%	28%	34%	37%	Strong growth
Go	8%	19%	19%	18%	Plateaued after growth
Rust	--	7%	9%	11%	Steady growth
PHP	30%	27%	20%	17%	Significant decline
Kotlin	2%	17%	16%	14%	Plateaued

Signal: JavaScript is still most used at 61%, but Python’s 32% → 57% climb and TypeScript’s 12% → 37% rise show the top-5 stack reshuffling.

Python Is Now the #1 Primary Language at 35%



Primary language

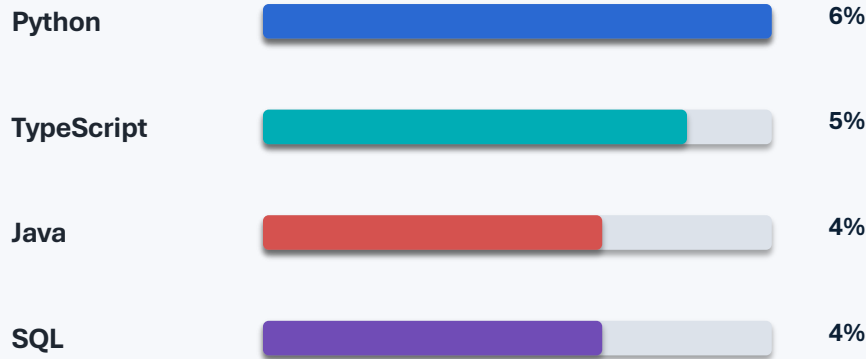


Primary language concentration

Python leads primary-language share at 35%, ahead of Java at 30% and JavaScript at 28%.

Adoption intent remains modest

Developers rarely plan to switch their primary stack; new-language adoption is

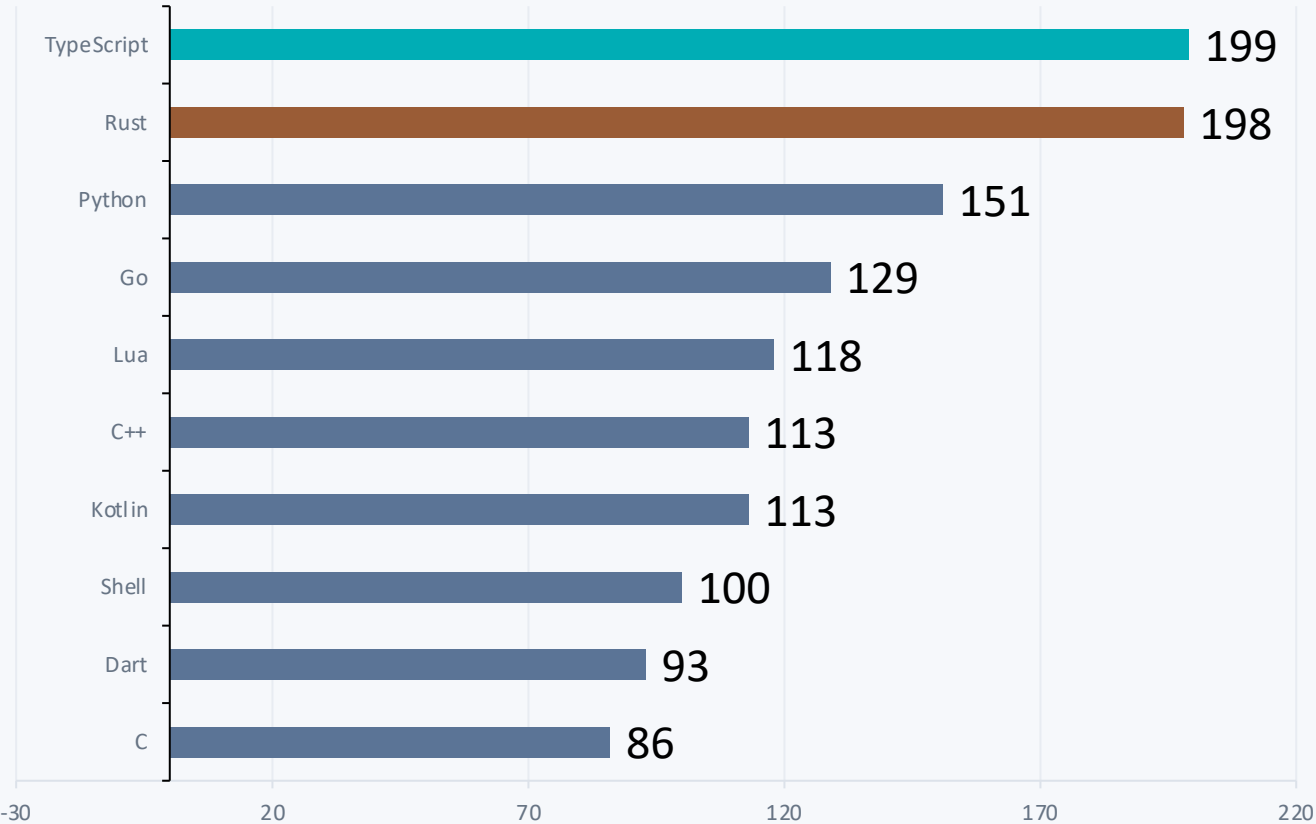


Implication: plan migrations around ecosystem lock-in, not just technical merits

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Satisfaction score



Satisfaction pattern

TypeScript leads with 199, just edging Rust at 198. Python follows with 151.

Why leaders should care

The strongest satisfaction signal clusters around type safety, code quality, and developer experience.

Retention lens: language ergonomics are part of developer experience.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Dominant language by application area

Application domain	Leading language	Share	Cluster	Strategic read
AI / Machine Learning	Python	31%	AI/ML	Invest in Python depth
Data Science / Analytics	Python	28%	Data	Standardize data stack
Data Processing & Analytics	Python	17%	Data	Standardize data stack
User Interface	TypeScript	62%	Frontend	Scale TS/JS skills
User Interface	JavaScript	58%	Frontend	Scale TS/JS skills
APIs & Services	Java	49%	Backend	Keep Java platform strong
Application Logic & Workflows	Java	55%	Backend	Keep Java platform strong
System Tools & Components	C++	27%	Systems	Specialist C++ hiring
Integrating with APIs & Services	Python	38%	Integration	Python glue layer

Python owns AI/data

31% AI/ML, 28% data science, 17% data processing, and 38% API integration

TypeScript owns UI

TypeScript leads UI at 62%, with JavaScript also dominant at 58%.

Java + C++ specialize

Java leads APIs/services and app workflows; C++ leads systems, hardware, a

Takeaway: language choice is driven by domain-fit more than

What This Means for Your Technology Strategy in 2025

Use language strategy as a portfolio decision: align stack choice, hiring, enablement, and retention with where each ecosystem is strongest.

1

Invest in Python for AI/ML and data teams

Python is the leading choice for AI/ML, data science, data processing, and integration use cases.

2

Treat TypeScript as the frontend/full-stack default

It leads UI usage and tops satisfaction, making it a strong developer-experience bet.

3

Use Go and Rust as infrastructure growth plays

Go stabilized after rapid growth; Rust continues climbing and both fit modern infrastructure needs.

4

Manage legacy-web exposure deliberately

PHP is in significant decline, while Ruby has dropped from 10% to 4%.

5

Make tooling part of retention strategy

Satisfaction leaders point to type safety, modern tooling, and stronger compiler guarantees.